

What does India's future look like?: Investigating India's fertility rate with the use of public data

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Introduction

This project investigated India's fertility rate variation over the years, specifically focusing on the question **"How has India's global fertility rate changed from 1950 to 2023, and how can that trend be used to predict rates for 2050?"**

India is currently the most populous country in the world (Worldometer,n.d.) [3], yet it is experiencing very low fertility rates. Experts believe this to be due to better education, contraceptives, higher costs of having a child, and the decrease in infant mortality rates (Shankar,2026) [2]. In 2023, India reached a fertility rate of 1.9 births per woman, which was below the benchmark of 2.1, meaning that it had crossed the point needed to keep the population stable (Shankar, 2026) [2]. It is crucial to stabilize this decline, as in 2005 India experienced a demographic dividend, where most of their population was in the workforce. With the low fertility rates, India will not be able to maintain the benefits of this dividend. The working population will age, and the population size will decrease, which will negatively affect their labour force.

Data Sample

The data was collected from "Our World in Data" [1], where 19,402 rows of data were downloaded. The results and filtered to include only data for India. The data analyzed was organized into 4 columns: entity, country's code, year, and fertility rate. The finalized set included 73 rows, not including the header.

Results and Discussion

The data from 1950-2023 was plotted into a scatterplot with a line of best fit to decipher an equation that best represents the relationship between the years and the fertility rate. The equation that was found was:

$$y = -0.0636x + 6.5524$$

To further develop the answer to the question, the data was back-tested, with the values from 1950 to 2020, to investigate whether it could predict the value for 2023. This gave the second equation of $y = -0.0637x + 6.5546$. The difference between the predicted value and the actual value for 2023 was 0.071, making it off by just 3.6%. As the back-test gave similar results for 2023, the prediction for 2050 was done using the original equation, which gave a value of 0.192 births per woman.

$$y = -0.0636(100) + 6.5524$$

$$y = 0.192 \text{ births per woman}$$

Furthermore, random sub-sampling was used, separating the data into 5 subgroups, each with around 14 points of data. A line of best fit was produced for each group, as visualized as blue lines in Figure 1. These regression lines were extrapolated for 2050 to calculate the overall mean and standard deviation.

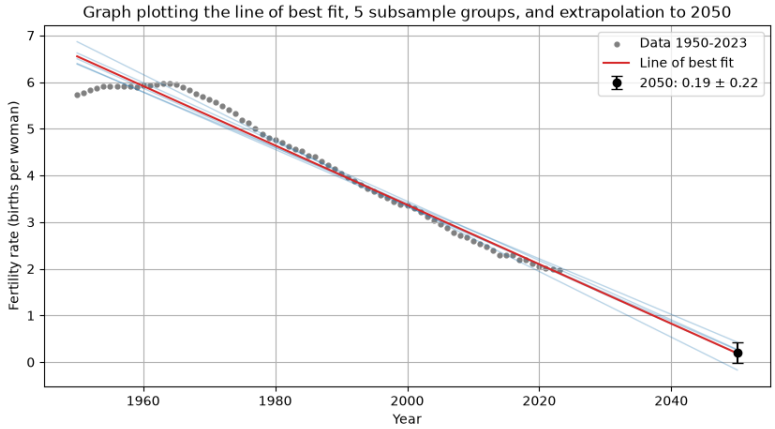


Figure 1: Scatterplot showcasing the forecast for 2050 fertility rates with regard to the 5 sub-sampling groups

When averaging the predictions from the sub-sampling groups for 2050, it gave a value of 0.19 +/- 0.22, where the 0.19 represents the average births per woman, and the 0.22 is the uncertainty value. The sub-sampling forecast closely agrees with the intial 2050 prediciton of 0.192 births per woman. This is evident in the figure, where the red line passes through the 2050 predicted value shown in black, as well as the values only being 0.002 off. The same data from the subgroups was then plotted into a Gaussian distribution to visually express where the data samples fit around the average point.

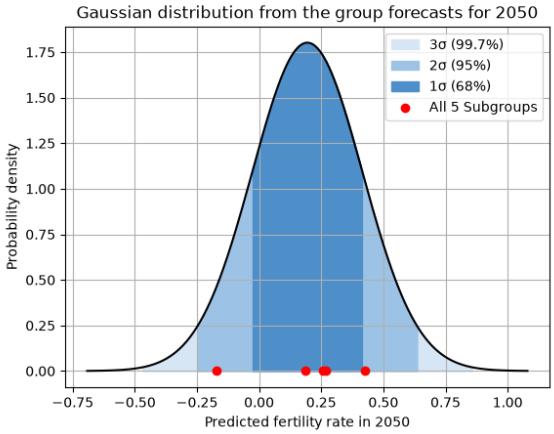


Figure 2: Gaussian distribution presenting subgroups placement in regards to average 2050 forecast

Figure 2 shows the five subgroup forecasts for 2050 against a Gaussian distribution with mean 0.19 and standard deviation 0.22. Three subgroups, with forecasts 0.185, 0.258, and 0.270, lie within the 1-sigma range (-0.03 to 0.42). The other two subgroups with the forecasts 0.426 and -0.170, lie within the 2-sigma range (-0.25 to 0.63). Nonetheless, it is crucial to note that the uncertainty value, 0.22, demonstrates the sensitivity to the data sampled, and this value would alter based on the different values of the sub-sampling groups if other numbers were used.

Overall the linear trend for 1950-2023 predicted a value of 0.192 births per woman for 2050, with a strong negative correlation being present. The sub-sampling technique produced a similar forecast, with a value of 0.19 and an uncertainty of 0.22. Nonetheless, the results achieved are not realistic, as the values for fertility rates may not drop into the negatives, yet one of the subgroups' forecasts was -0.170. To expand on this investigation, a different model, such as a logistic function, could be used to predict the value for 2050. Additionally, the uncertainty value described the variation of the linear model and was based on the numbers chosen randomly by the sub-samples, reducing its reliability for the uncertainty value. Further random sub-sampling and repetition of the experiment would benefit the investigation by providing a more stable sigma value.

References

[1] Our World in Data. (2025). Fertility rate: Births per woman. Retrieved September 20, 2026, from https://ourworldindata.org/grapher/children-born-per-woman?tab=table&tableSearch=india.

[2] Shankar, P. (2026, June 9). India's fertility rate falls below replacement level: Why it matters. Al Jazeera. https://www.aljazeera.com/news/2026/6/9/indias-fertility-rate-falls-below-replacement-level-why-it-matters .

[3] Worldometer. (n.d.). Countries in the world by population (2026). Retrieved October 8, 2026, from https://www.worldometers.info/world-population/population-by-country/ .